MENTOR CALL

JavaScript Programming

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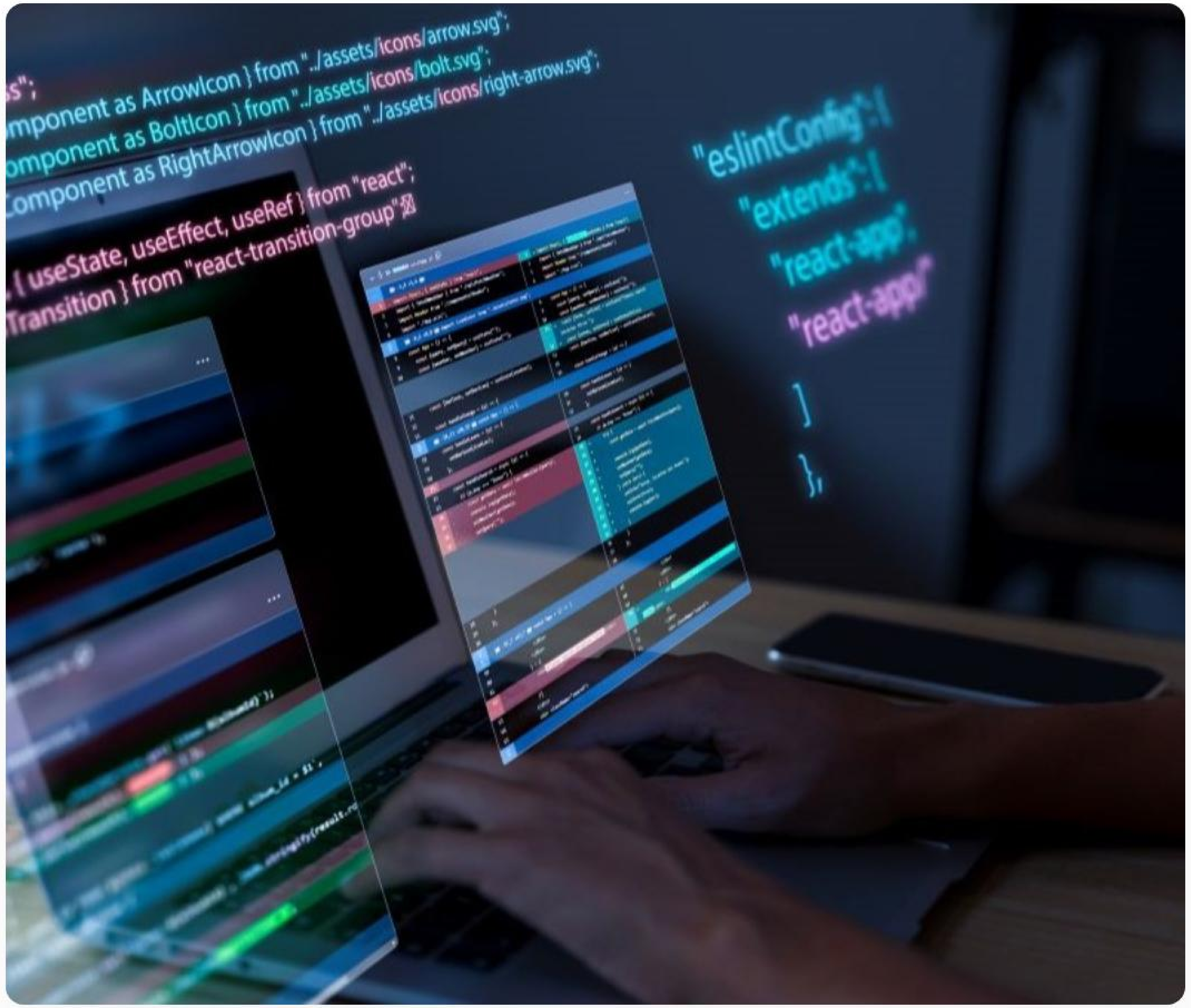


ZINDUA SCHOOL

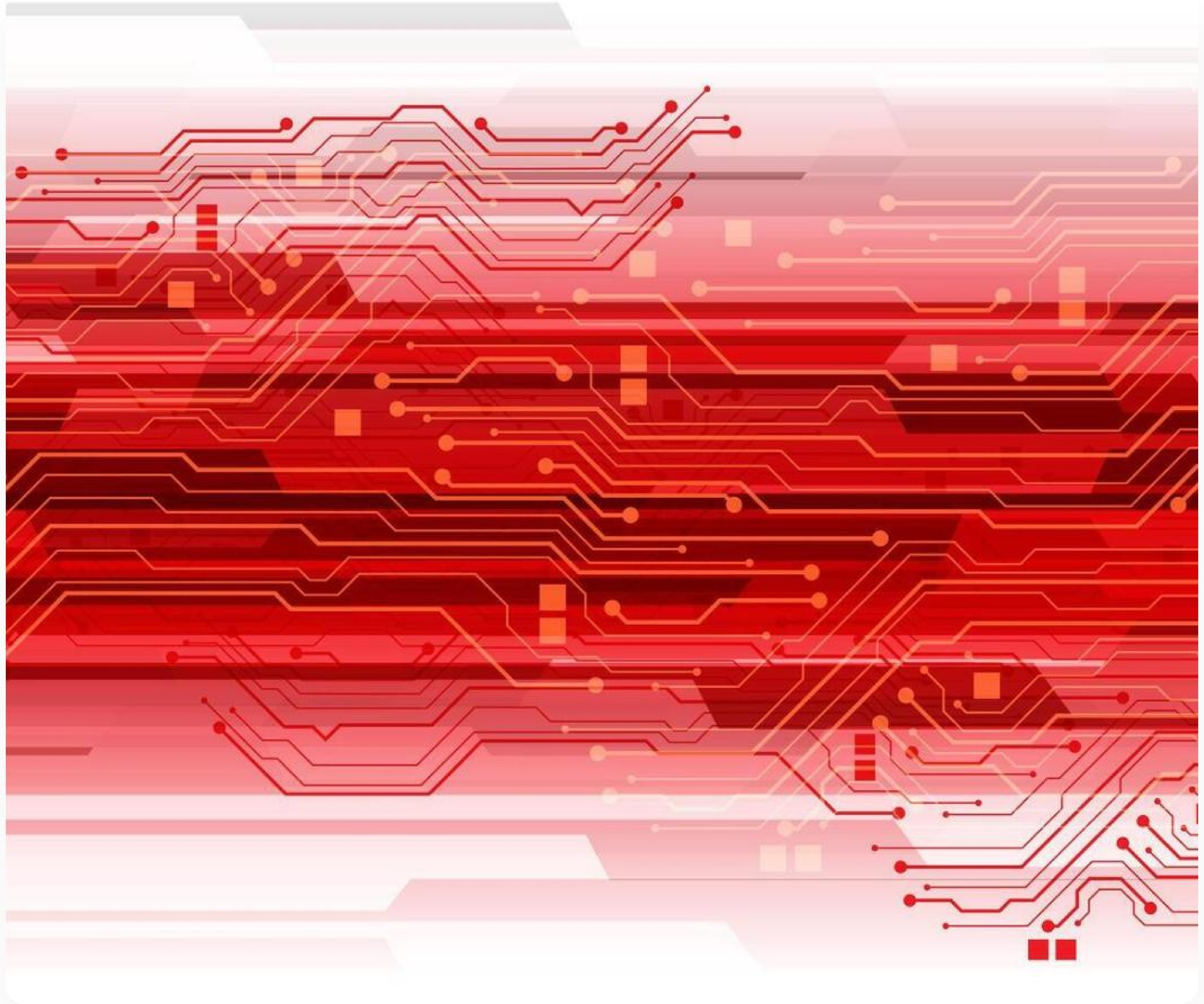
01. Foundations

Variables, Types, and Memory

What is JavaScript?

JavaScript is a **high-level, interpreted** programming language that is a core technology of the World Wide Web.

- **Dynamic:** Adds interactivity to static pages.
- **Universal:** Runs on Client & Server.
- **Prototypal:** Object-based architecture.



Variables: Declarations

const

For values that **never change**. Block-scoped and the preferred modern standard.

```
const id = 101;
```

let

For variables that **need reassignment**. Also block-scoped, solving old scope bugs.

```
let score = 0;
```

var

The legacy declaration. Function-scoped. **Not recommended** for modern development.

```
var x = 5;
```

Primitive Data Types

String

Text data wrapped in quotes.

`"Hello World"`

Number

Integers or decimals.

`42` or `3.14`

Boolean

Logical values.

`true` or `false`

Null

Intentional absence of value.

`null`

Code Comments

Comments help developers communicate intent and logic without affecting execution.

- **Single-line:** Use `//` for quick notes.
- **Multi-line:** Use `/* */` for blocks.

Type Conversion

Input Type	Target Type	Method	Example Result
String	Number	`Number("12")`	12
Number	String	`String(45)`	"45"
Boolean	Number	`Number(true)`	1
String	Boolean	`Boolean("Hi")`	true

Implicit Coercion

"5" + 2

Becomes "52"

"5" - 2

Becomes 3

JavaScript coercion converts types automatically during operations, which can lead to unexpected results.

Arithmetic Operators

- ➕ Addition: `+`
- ➖ Subtraction: `-`
- ✱ Multiplication: `*`
- ➗ Division: `/`
- ٪ Modulo (Remainder): `%`
- ^ Exponent: `**`

Comparison Operators

=

Strict Equality

`===` compares both **value** and **type**. Always use this.

≠

Inequality

`!==` checks if values or types are different.

≥

Relational

`>`, `<`, `>=`, `<=` used for numeric logic.

Collaborative Learning

Ready to Practice?

Programming is a craft learned through repetition and experimentation.

- Open your browser console.
- Try declaring a `const`.
- Test implicit coercion.

Questions?

HAPPY CODING!

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zindua.io/curriculum

Image Sources

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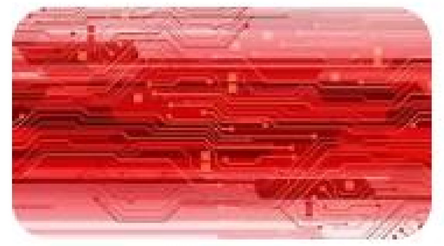

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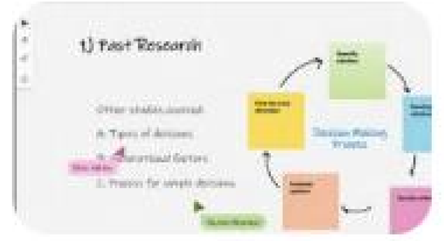

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